

Solubility Prediction Step 2

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Importing libraries:

```
library(tidyverse)
library(kableExtra)
```

Reading the data file;

```
df <- read.csv("df.csv", sep = ",", header = TRUE)
```

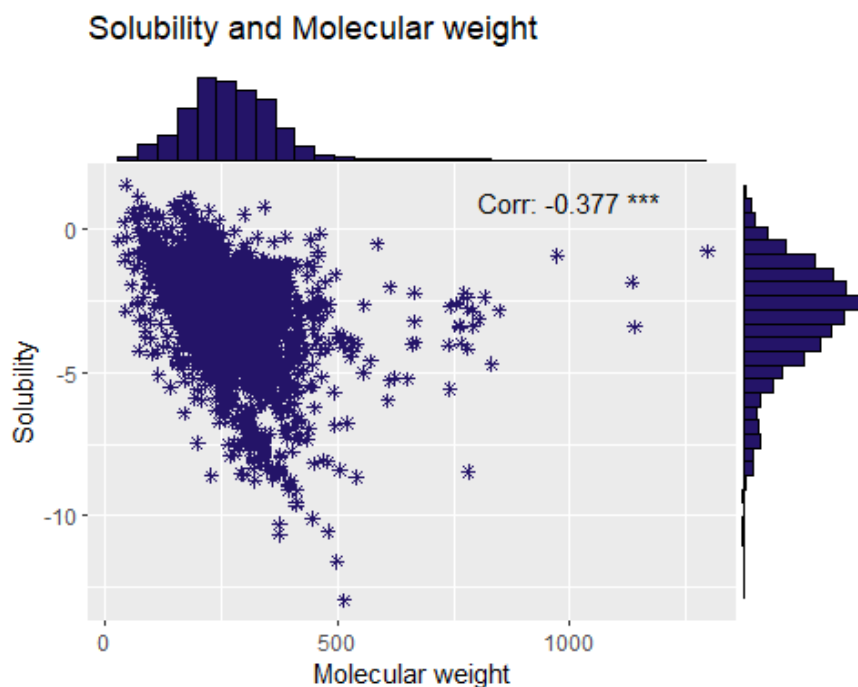
Making the correlation values:

```
S_mw = round(cor(df$Solubility, df$MolWt, use = "complete.obs"), digits = 3)
S_mLP = round(cor(df$Solubility, df$MolLogP, use = "complete.obs"), digits = 3)
S_HBA = round(cor(df$Solubility, df$NumHAcceptors, use = "complete.obs"), digits = 3)
S_HBD = round(cor(df$Solubility, df$NumHDonors, use = "complete.obs"), digits = 3)
```

Basic scatter plot of solubility and Lipinski specific descriptors:

```
scatter_MW <- {scatter_plot_1 <- ggplot(df) +
  geom_point(shape = 8, color = "#241468", mapping = aes(y = Solubility,
    x = MolWt)) +
  labs(title = "Solubility and Molecular weight") + xlab('Molecular weight') +
  annotate("text", x = 1000, y = 1, label = paste("Corr:", S_mw, '***'))
ggExtra::ggMarginal(
  scatter_plot_1,
  type = "histogram",
  fill = "#241468")
}
```

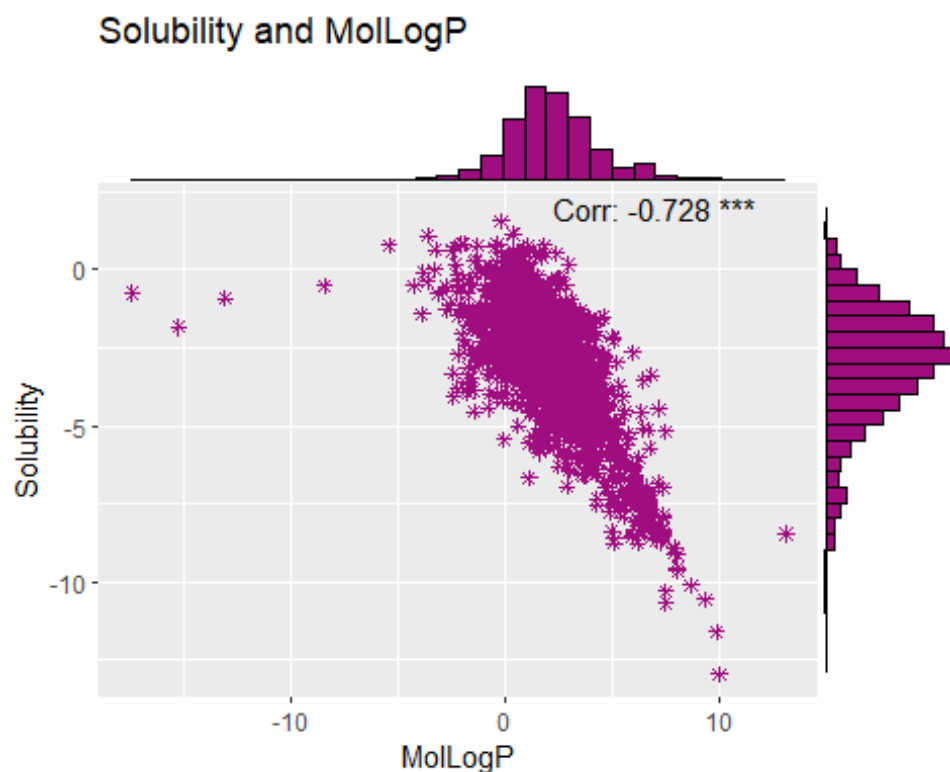
scatter_MW



```
scatter_M1 <- {scatter_plot_3 <- ggplot(df) +
  geom_point(shape = 8,color = "#9F0D7F", mapping = aes(y = Solubility,
                                                         x = MolLogP)) +

  labs(title = "Solubility and MolLogP") +
  annotate("text", x = 7, y = 2, label = paste("Corr:", S_mLP, '***'))
ggExtra::ggMarginal(
  scatter_plot_3,
  type = "histogram",
  fill = "#9F0D7F")
}
```

scatter_M1

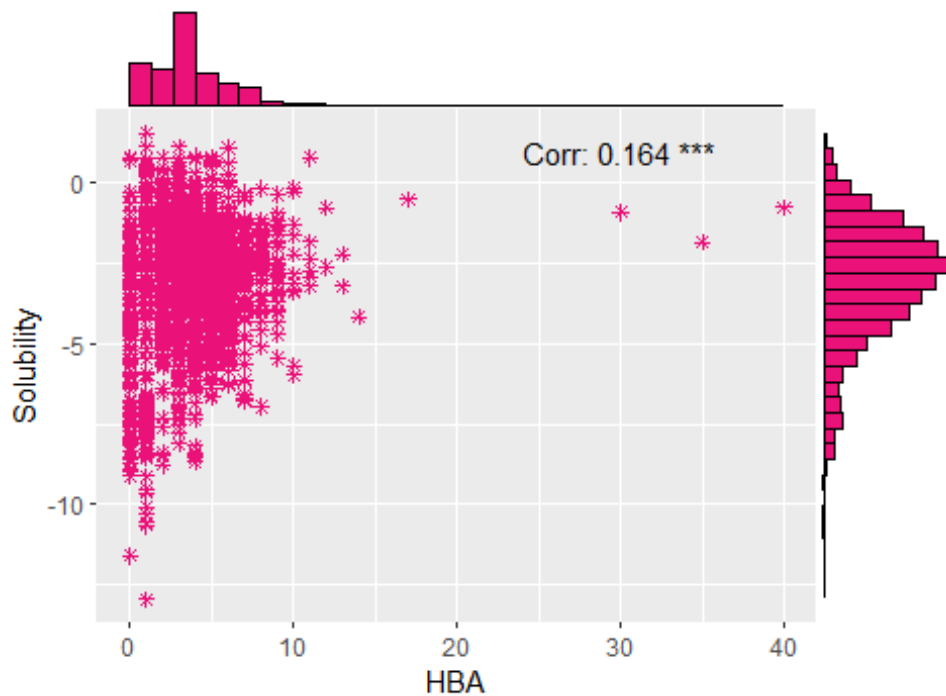


```
scatter_HBA <- {scatter_plot_3 <- ggplot(df) +
  geom_point(shape = 8,color = "#EA1179", mapping = aes(y = Solubility,
                                                         x = NumHAcceptors)) +

  labs(title = "Solubility and HBA") + xlab('HBA') +
  annotate("text", x = 30, y = 1, label = paste("Corr:", S_HBA, '***'))
ggExtra::ggMarginal(
  scatter_plot_3,
  type = "histogram",
  fill = "#EA1179")
}
```

scatter_HBA

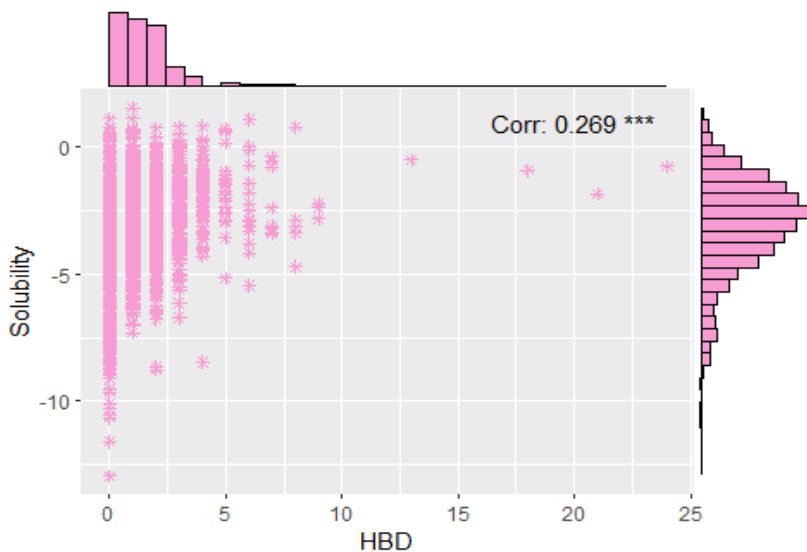
Solubility and HBA



```
scatter_HBD <- {scatter_plot_3 <- ggplot(df) +
  geom_point(shape = 8,color = "#F79BD3", mapping = aes(y = Solubility,
                                                         x = NumHDonors)) +
  labs(title = "Solubility and HBD") + xlab('HBD') +
  annotate("text", x = 20, y = 1, label = paste("Corr:", S_HBD, '***'))
ggExtra::ggMarginal(
  scatter_plot_3,
  type = "histogram",
  fill = "#F79BD3")
}
```

scatter_HBD

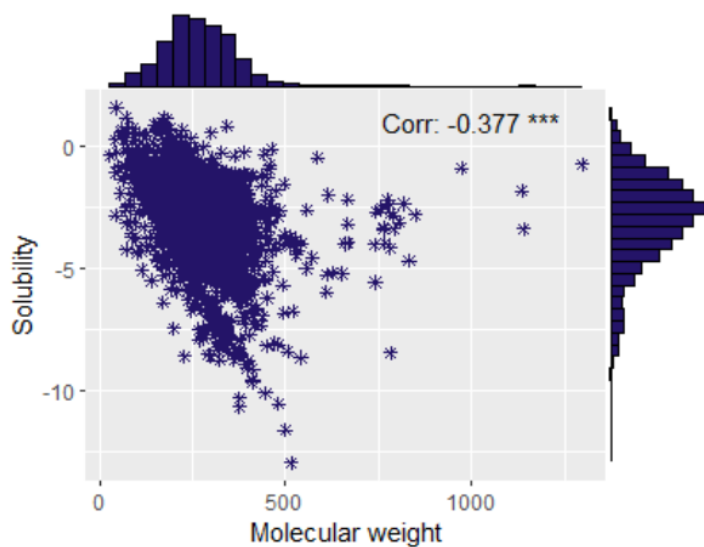
Solubility and HBD



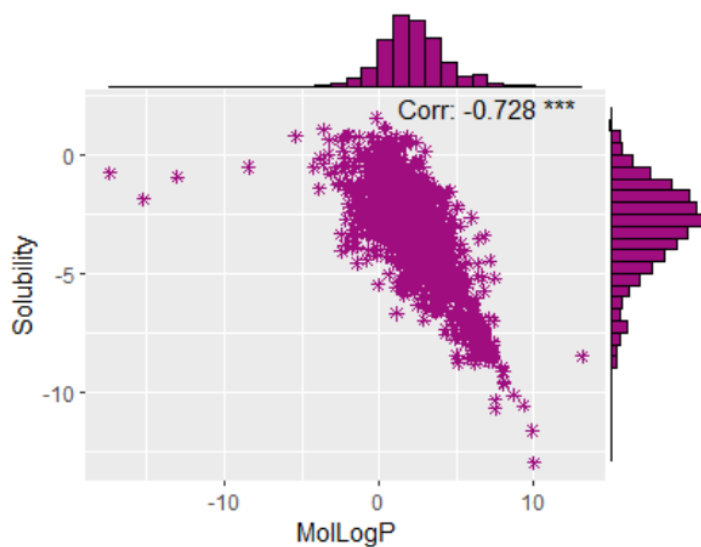
```
plot_grob <- gridExtra::arrangeGrob(grobs = list(scatter_MW, scatter_ML,
                                                  scatter_HBA, scatter_HBD))
gridExtra::grid.arrange(plot_grob,
                        top = paste("Correlation plots of Lipinski parameters"))
```

Correlation plots of Lipinski parameters

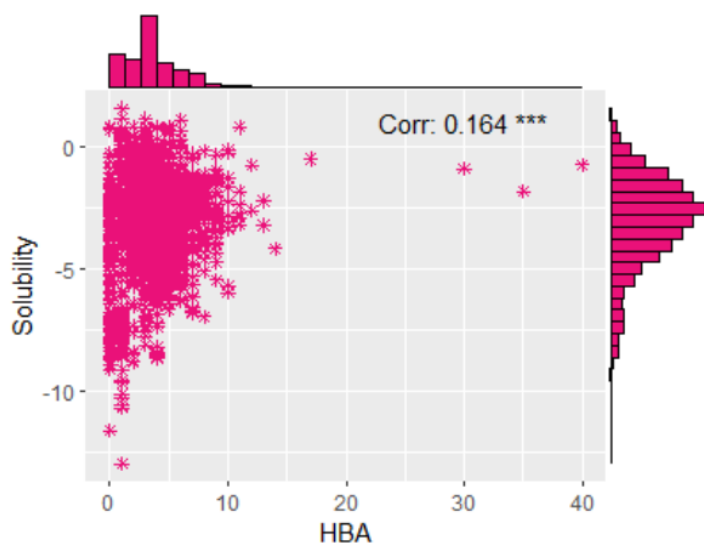
Solubility and Molecular weight



Solubility and MolLogP



Solubility and HBA



Solubility and HBD

